MAPS FROM THE SPECTRUM OF A COMPLETE LOCAL NOETHERIAN ALGEBRA AGREE WITH MAPS FROM ITS FORMAL SPECTRUM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The last sentence of the proof is the author's 'We omit the proof that x_n is the restriction of x to Spec(R/m^n).' It is reproduced verbatim and is recorded as an obligation of the card; both the injectivity and the surjectivity of the displayed map are proved in full inside the counted span.

Editorial note. The proof displays four commutative diagrams; all lie inside the counted span and are reproduced verbatim.

Editorial note. The counted spans carry 8 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Morphisms into a formal spectrum as an inverse limit over the quotients. Same chapter, the identification used in the first line of the proof at proof lines 7240-7244. Turns the asserted bijectivity into a statement about the inverse limit of the morphism sets over the quotients of R.
- Properties of the diagonal of an algebraic space, and quasi-finite morphisms of finite type are locally Noetherian. Morphisms of Spaces, Lemma spaces-morphisms-lemma-properties-diagonal, invoked at proof lines 7249-7251, and Morphisms, Lemma morphisms-lemma-finite-type-noetherian at proof lines 7251-7253. Gives that the fibre product with the diagonal is a locally Noetherian scheme over the spectrum of R.
- A Noetherian local ring embeds in its completion. Algebra, Lemma algebra-lemma-intersect-powers-ideal-module-zero, invoked at proof lines 7259-7260. Upgrades the agreement of all the quotients to the equality of the local ring with R, which gives injectivity.
- The fibre of an etale morphism over a field, finite etale extensions of complete local rings, and formal etaleness. Morphisms, Lemma morphisms-lemma-etale-over-field at proof lines 7273-7275, Algebra, Lemmas algebra-lemma-henselian-cat-finite-etale and algebra-lemma-complete-henselian at

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proof lines 7286-7288, and More on Morphisms of Spaces, Lemma spaces-more-morphisms-lemma-etale-formally-etale at proof lines 7290-7293. Supplies the finite separable extension and the lifting of the compatible family along the etale cover, which is the heart of the surjectivity argument.
- The same statement for schemes. Same chapter, Lemma lemma-map-into-scheme, invoked at proof lines 7294-7295. Converts the lifted compatible family on the cover into a morphism from the spectrum of the extended complete local ring.

Source reference. Original lemma 04GM: sources/stacks-chow/tags.txt, source lines 5838–5838. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04GK: sources/stacks-chow/tags.txt, source lines 5836–5836. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00IP: sources/stacks-chow/tags.txt, source lines 687–687. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AQG: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7203–7203. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0616: sources/stacks-chow/tags.txt, source lines 7750–7750. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03HK: sources/stacks-chow/tags.txt, source lines 4570–4570. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let R be a complete local Noetherian S-algebra. Let X be an algebraic space over S. Then the natural map*

$$\mathrm{Mor}_S(\mathrm{Spec}(R), X) \longrightarrow \mathrm{Mor}_S(Spf(R), X)$$

is bijective.

Proof. Let $\mathfrak{m}$ be the maximal ideal of R. We have to show that

$$\mathrm{Mor}_S(\mathrm{Spec}(R), X) \longrightarrow \lim \mathrm{Mor}_S(\mathrm{Spec}(R/\mathfrak{m}^n), X)$$

is bijective for R as above.

Injectivity: Let $x, x' : \mathrm{Spec}(R) \to X$ be two morphisms mapping to the same element in the right hand side. Consider the fibre product

$$T = \mathrm{Spec}(R) \times_{(x,x'),X\times_S X,\Delta} X$$

Then T is a scheme and $T \to \mathrm{Spec}(R)$ is locally of finite type, monomorphism, separated, and locally quasi-finite, see Morphisms of Spaces, Lemma 03HK. In particular T is locally Noetherian, see Morphisms, Lemma 01T6. Let $t \in T$ be the unique point mapping to the closed point of $\mathrm{Spec}(R)$ which exists as x and x' agree over $R/\mathfrak{m}$. Then $R \to \mathcal{O}_{T,t}$ is a local ring map of Noetherian rings such that $R/\mathfrak{m}^n \to \mathcal{O}_{T,t}/\mathfrak{m}^n\mathcal{O}_{T,t}$ is an isomorphism for all n (because x and x' agree over $\mathrm{Spec}(R/\mathfrak{m}^n)$ for all n). Since $\mathcal{O}_{T,t}$ maps injectively into its completion (see Algebra, Lemma 00IP) we conclude that $R = \mathcal{O}_{T,t}$. Hence x and x' agree over R.

Surjectivity: Let (x_n) be an element of the right hand side. Choose a scheme U and a surjective étale morphism $U \to X$. Denote $x_0 : \mathrm{Spec}(k) \to X$ the morphism induced on the residue field $k = R/\mathfrak{m}$. The morphism of schemes $U \times_{X,x_0} \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is surjective étale. Thus $U \times_{X,x_0} \mathrm{Spec}(k)$ is a nonempty disjoint union of spectra of finite separable field extensions of k, see Morphisms, Lemma 02GL. Hence we can find a finite separable field extension k'/k and a k'-point $u_0 : \mathrm{Spec}(k') \to U$ such that

$$\begin{array}{ccc} \mathrm{Spec}(k') & \xrightarrow[u_0]{} & U \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \xrightarrow{x_0} & X \end{array}$$

commutes. Let $R \subset R'$ be the finite étale extension of Noetherian complete local rings which induces k'/k on residue fields (see Algebra, Lemmas 04GK and 04GM). Denote x'_n the restriction of x_n to $\mathrm{Spec}(R'/\mathfrak{m}^n R')$. By More on Morphisms of Spaces, Lemma 0616 we can find an element $(u'_n) \in \lim \mathrm{Mor}_S(\mathrm{Spec}(R'/\mathfrak{m}^n R'), U)$ mapping to (x'_n). By Lemma 0AQG the family (u'_n) comes from a unique morphism $u' : \mathrm{Spec}(R') \to U$. Denote $x' : \mathrm{Spec}(R') \to X$ the composition. Note that $R' \otimes_R R'$ is a finite product of spectra of Noetherian complete local rings to which our current discussion applies. Hence the diagram

$$\begin{array}{ccc} \mathrm{Spec}(R' \otimes_R R') & \longrightarrow & \mathrm{Spec}(R') \\ \downarrow & & \downarrow{\scriptstyle x'} \\ \mathrm{Spec}(R') & \xrightarrow{x'} & X \end{array}$$

is commutative by the injectivity shown above and the fact that x'_n is the restriction of x_n which is defined over $R/\mathfrak{m}^n$. Since $\{\mathrm{Spec}(R') \to \mathrm{Spec}(R)\}$ is an fppf covering we conclude that x' descends to a morphism $x : \mathrm{Spec}(R) \to X$. We omit the proof that x_n is the restriction of x to $\mathrm{Spec}(R/\mathfrak{m}^n)$. □